Chapter 4: Energy

1.

$$\frac{d}{dt}(\vec{a}\circ\vec{b}) = \frac{d}{dt}(a_x \cdot b_x + a_y \cdot b_y)$$

$$= \frac{d}{dt}(a_x b_x) + \frac{d}{dt}(a_y \cdot b_y)$$

$$= \frac{da_x}{dt} b_x + a_x \frac{db_x}{dt} + \frac{da_y}{dt} b_y + a_y \frac{db_y}{dt}$$

$$= \frac{d\vec{a}}{dt}\circ\vec{b} + \vec{a}\circ\frac{d\vec{b}}{dt}$$

2.

(from problem) "Work"

$$W = \int_0^P F\circ dr = \int_0^P (F_x dx + F_y dy) \quad , \quad \vec{F} = (x^2, 2xy)$$

a) $O = (0,0)$, $P = (1,1)$

$$W = \int_0^1 x^2 dx + \int_0^1 2xy\, dy \qquad @\, x = 1$$

$$= 1/3 + 1$$

$$= 4/3$$

b) (From problem) "Term replaced"

$$dy = 2x\,dx$$

$$W = \int_0^1 x^2 dx + \int_0^1 2xy(2x\,dx)$$

$$= 1/3 + \int_0^1 4x^2 \cdot y\,dx \quad @y=1$$

$$= 1/3 + 4/3$$

$$= 5/3$$

c) (from problem) "Parametric terms"

$$x = t^3, \quad y = t^2$$

$$dx = 3t^2 dt \quad dy = 2t\,dt$$

$$W = \int_0^1 t^6 (3t^2 dt) + \int_0^1 2(t^3)(t^2)(2t\,dt)$$

$$= 1/3 + 4/7$$

$$= 7/21 + 12/21$$

$$= 19/21$$

3. (4.100) "Work"

$$W = \int_O^P F \cdot dr = \int_O^P (F_x dx + F_y dy)$$

$$= \int_O^Q (F_x dx + F_y dy) + \int_Q^P (F_x dx + F_y dy)$$

(from problem) "Force"

$$F = (-y, x)$$

a) $P = (1,0)$, $Q = (0,1)$, $O = (0,0)$, $P \to O \to Q$

$$W = \int_{P=(1,0)}^{O=(0,0)} -y\,dx + x\,dy + \int_{O=(0,0)}^{Q=(0,1)} -y\,dx + x\,dy$$

$$= -\int_0^1 y\,dx + \int_0^0 x\,dy - \int_0^0 y\,dx - \int_0^1 x\,dy$$

$$= 0$$

b) $P = (1,0)$, $Q = (0,1)$, $O = (0,0)$ $P \to Q$

Also, $y = -x + 1$, $-dy = dx$

...

$$W = \int_{P=(1,0)}^{Q=(0,1)} -y\,dx + x\,dy$$

$$= \int_{P=(1,0)}^{Q=(0,1)} (x-1)\,dx - x\,dx$$

$$= -\int_{1}^{0} dx$$

$$= \underline{1}$$

c) $P=(1,0),\ Q=(0,1)$; $P \to Q$

Also, $x = \cos\phi$, $y = \sin\phi$, $dx = -\sin\phi\,d\phi$

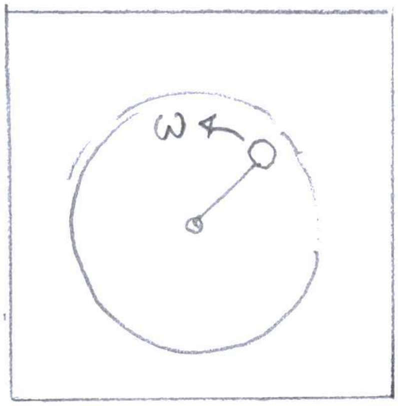
$dy = \cos\phi\,d\phi$

$$W = \int_{P=(1,0)}^{Q=(0,1)} -y\,dx + x\,dy$$

$$= \int_{P=(1,0)}^{Q=(0,1)} +\sin^2\phi\,d\phi + \cos^2\phi\,d\phi$$

$$= \int_{0}^{\pi/2} d\phi = \pi/2$$

4.



"Particle... on a table... attached to a string."

a) Angular Velocity:

(3.18) "Angular Momentum"

$$l = mr^2\omega$$

Angular Momentum Initially
= Angular Momentum Finally

$$l_i = l_f$$

$$mr_0^2\omega_0 = mr^2\omega$$

$$\omega = \left(\frac{r_0}{r}\right)^2 \omega_0$$

b) (4.7) "Work-KE Theorem"

$$\Delta T = T_2 - T_1 = \int_1^2 F \cdot dr \equiv W(1 \to 2)$$

$$W = \int_{r_0}^{r} F \cdot dr$$

$$= \int_{r_0}^{r} mr\omega^2 dr$$

$$= \int_{r_0}^{r} mr\left(\frac{r_0^2}{r}\right)^4 \omega_0^2 dr$$

$$= m r_0^4 \omega_0^2 \int_{r_0}^{r} \frac{1}{r^3}\, dr$$

$$= \frac{m r_0^4 \omega_0^2}{2}\left(\frac{1}{r_0^2} - \frac{1}{r^2}\right)$$

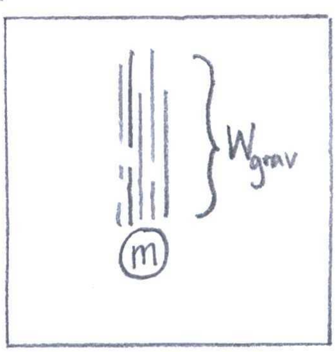
Absolute Work $= |W|$

c) Kinetic Energy:

$$\Delta T = W(1 \to 2)$$

$$= \frac{m r_0^2 \omega_0^2}{2}\left(\frac{1}{r^2} - \frac{1}{r_0^2}\right)$$

5.



"mass moves... by gravity"

a) Work by gravity:

$$W = F \cdot \Delta y$$

$$= mg \cdot (y_{lower} - y_{higher})$$

$$= -mg \cdot h \quad \ldots \text{ when } \Delta y = \text{height}(h)$$

b) Gravitational Potential Energy:

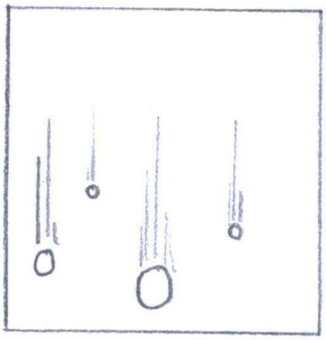
(4.54) "Potential Energy"

$$U(x) = -\int_{x_0}^{x} F_x(x')\, dx$$

$$= -\int_{y_{higher}}^{y_{lower}} mg\,dy$$

$$= mgy$$ "constant is conservative"

6.



"particles subject to a uniform gravitational field."

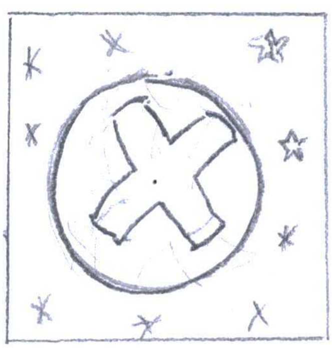
$$U_{TOTAL} = U_1 + U_2 + U_3 + \cdots + U_N$$

$$= \sum_{\alpha=1}^{N} U_\alpha$$

$$= \sum_{\alpha=1}^{N} m_\alpha \cdot g \cdot Y$$

$$= MgY \quad \text{where} \quad M = \sum_{\alpha=1}^{N} m_\alpha$$

7.



"Planet X"

a) <u>Work by Gravity:</u>

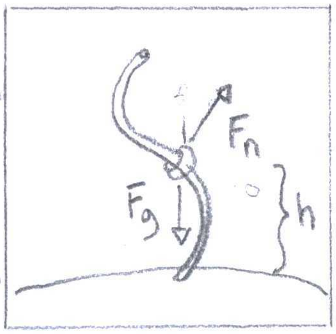
$$W = \int_{r_1}^{r_2} F \circ dr \quad \ldots (4.10)$$

$$= \int_{r_1}^{r_2} m\gamma r^2 dr$$

$$= \frac{m\gamma}{3}\left[r_2^3 - r_1^3\right]$$

<u>Potential energy:</u>

$$U = -W = \frac{m\gamma}{3}\left[r_2^3 - r_1^3\right] \quad \ldots (4.13)$$

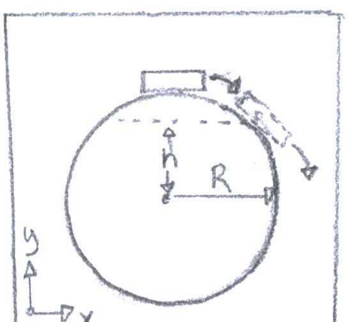


b) The total potential energy is constant. In force units, gravity is conservative. Net total energy (or force) depends on initial height, no matter the wire shape, dimension, or length.

c) Velocity:

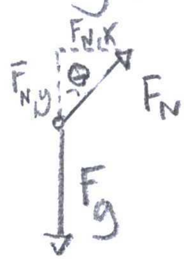
$$V = V_0 + at$$
$$= -m\gamma h^2 \cdot t$$

8.



"Puck perched at the top of a fixed Sphere."

Free-body Diagram:



Height(h) puck leaves sphere:

(4.20) "Mechanical Energy"

$$E = T + U$$
$$= T_i + U_i$$
$$= T_f + U_f$$

$$= mgR$$

$$= \frac{1}{2}mv^2 + mgR\cos\theta$$

$$\frac{1}{2}mv^2 = mgR(1-\cos\theta)$$

$$\vec{F}_a = m\frac{v^2}{R}$$

$$= mg\cos\theta - \vec{F}_N$$

$$\vec{F}_N = mg\cos\theta - \frac{mv^2}{R}$$

$$= mg\cos\theta - \frac{2mgR(1-\cos\theta)}{R}$$

$$= mg(3\cos\theta - 2)$$ ... at $\theta = 0°$, normal force is mg

$$= 0$$ ... when puck leaves sphere

$$\theta = \cos^{-1}\left(\frac{2}{3}\right)$$

$$\cong 48°$$

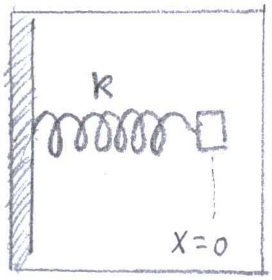
$$h = R\cos\theta$$

$$= (2/3)R$$

9.

a) (From problem) "Force by spring"

$$F = -kx$$



"one-dimensional spring"

Potential energy from a spring:

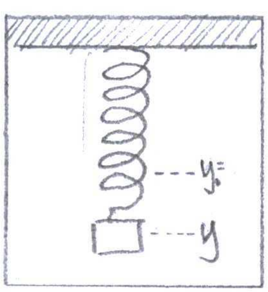
(4.13) "Potential energy"

$$U(r) = -W(r_0 \to r) \equiv -\int_{r_0}^{r} F(r') \cdot dr'$$

$$\cong -\int_0^x (-kx) \cdot dx$$

$$\cong \frac{1}{2} k x^2$$

b) Extension from y_0:



"spring... hung vertically"

Free-body diagram:

Force: $F_s = -F_g$

$$-ky = -mg$$

$$y = \frac{mg}{k}$$

Total Potential Energy:

(4.13) "Potential Energy"

$$U(r) = -W(r_0 \to r) \equiv -\int_{r_0}^{r} F(r')\cdot dr'$$

$$\cong -\int_0^y (-ky)\,dy$$

$$\cong \frac{1}{2}ky^2$$

10. a) $f(x,y,z) = ax^2 + bxy + cy^2$

$$\left(\frac{df}{dx}\right)_{y,z} = 2ax + by$$

$$\left(\frac{df}{dy}\right)_{x,z} = bx + 2cy$$

$$\left(\frac{df}{dz}\right)_{x,y} = 0$$

b) $g(x,y,z) = \sin(axyz^2)$

$$\left(\frac{dg}{dx}\right)_{y,z} = \cos(axyz^2)\cdot ayz^2$$

$$\left(\frac{dg}{dy}\right)_{x,z} = \cos(axyz^2)\cdot 2axyz$$

$$\left(\frac{dg}{dz}\right)_{x,y} = \cos(axyz^2)\cdot axz^2$$

$$\boxed{\begin{array}{c}\text{Chain Rule} \\ \dfrac{dg(h(x))}{dx} = \dfrac{dg}{dx}\cdot\dfrac{dh(x)}{dx}\end{array}}$$

c) $h(x,y,z) = a e^{xy/z^2}$

$$\left(\frac{dh}{dx}\right)_{y,z} = \frac{ay}{z^2} e^{xy/z^2}$$

$$\left(\frac{dh}{dy}\right)_{x,z} = \frac{ax}{z^2} e^{xy/z^2}$$

$$\left(\frac{dh}{dz}\right)_{x,y} = -2\frac{axy}{z^3} e^{xy/z^2}$$

5.11 a) $f(x,y,z) = ay^2 + 2byz + cz^2$

$$\left(\frac{df}{dx}\right)_{y,z} = 0$$

$$\left(\frac{df}{dy}\right)_{x,z} = 2ay + 2bz$$

$$\left(\frac{df}{dz}\right)_{x,y} = 2by + 2cz$$

b) $g(x,y,z) = \cos(axy^2z^3)$

$$\left(\frac{dg}{dx}\right)_{y,z} = -ay^2z^3\sin(axy^2z^3)$$

$$\left(\frac{dg}{dy}\right)_{xz} = -2axyz^3\sin(axy^2z^3)$$

$$\left(\frac{dg}{dz}\right)_{x,y} = -3axy^2z^2\sin(axy^2z^3)$$

c) $h(x,y,z) = ar$ where $r = \sqrt{x^2 + y^2 + z^2}$

$$\left(\frac{dh}{dx}\right)_{y,z} = \frac{ax}{r}$$

$$\left(\frac{dh}{dy}\right)_{x,z} = \frac{ay}{r}$$

$$\left(\frac{dh}{dz}\right)_{x,y} = \frac{az}{r}$$

12.12

a) $f = x^2 + y^2$

$$\nabla f = \hat{x}\frac{df}{dx} + \hat{y}\frac{df}{dy} + \hat{z}\frac{df}{dz}$$

$$= 2x \cdot \hat{x} + 2y \cdot \hat{y}$$

$\boxed{\text{Back Cover}}$

Cartesian

Gradient

$\nabla f = \hat{x}\frac{df}{dx} + \hat{y}\frac{df}{dy} + \hat{z}\frac{df}{dz}$

b) $f = ky$ where k = constant

$$\nabla f = \hat{x}\frac{df}{dx} + \hat{y}\frac{df}{dy} + \hat{z}\frac{df}{dz}$$

$$= k\hat{y}$$

c) $f = r \equiv \sqrt{x^2 + y^2 + z^2}$

$$\nabla f = \hat{x}\frac{df}{dx} + \hat{y}\frac{df}{dy} + \hat{z}\frac{df}{dz}$$

$$= \left(\frac{x}{r}\right)\hat{x} + \left(\frac{y}{r}\right)\hat{y} + \left(\frac{z}{r}\right)\hat{z}$$

d) $f = 1/r$

$$\nabla f = \hat{x}\frac{df}{dx} + \hat{y}\frac{df}{dy} + \hat{z}\frac{dF}{dz}$$

$$= -\left(\frac{x}{r^3}\right)\hat{x} - \left(\frac{y}{r^3}\right)\hat{y} - \left(\frac{z}{r^3}\right)\hat{z}$$

13. a) $f = \ln(r)$ ... where $r = \sqrt{x^2 + y^2 + z^2}$

$$\nabla f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$= \frac{x\hat{x} + y\hat{y} + z\hat{z}}{r^2}$$

b) $f = r^n$

$$\nabla f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$= nr^{n-2}(x\hat{x} + y\hat{y} + z\hat{z})$$

c) $f = g(r)$ ... where $g(r)$ is unspecific

$$\nabla f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$= \frac{x\hat{x} + y\hat{y} + z\hat{z}}{r}\frac{\partial g}{\partial r}$$

14.

$$\nabla(fg) = \hat{x}\frac{\partial(fg)}{\partial x} + \hat{y}\frac{\partial(fg)}{\partial y} + \hat{z}\frac{\partial(fg)}{\partial z}$$

$$= \hat{x}\left(f\frac{\partial g}{\partial x} + g\frac{\partial f}{\partial x}\right) + \hat{y}\left(f\frac{\partial g}{\partial y} + g\frac{\partial f}{\partial y}\right) + \hat{z}\left(f\frac{\partial g}{\partial z} + g\frac{\partial f}{\partial z}\right)$$

$$= f\bar{\nabla} g + g\nabla f$$

15 (4.35) "Total Differential"

$$df = \nabla f \circ dr$$

$$= \frac{df}{dx}dx + \frac{df}{dy}dy + \frac{df}{dz}dz$$

$$\doteq f(x+dx) - f(x) + f(y+dy) - f(y) + f(z+dz) - f(z)$$

$\circ\circ\circ$ by (4.27) modified

(from problem) $f(r) = x^2 + 2y^2 + 3z^2$

and $r = (1,1,1)$ to $(1.01, 1.03, 1.05)$

$$df = (1.01)^2 - (1)^2 + 2(1.03)^2 - 2(1)^2 + 3(1.05)^2 - 3(1)^2$$

$$= 0.45$$

Comparison to Exact Result:

Exact : $f = \int_{r=(1,1,1)}^{r=(1.01,1.03,1.05)} \nabla f \circ dr$

$$= \int_1^{1.01} 2x\,dx + \int_1^{1.03} 4y\,dy + \int_1^{1.05} 6z\,dz$$

$$= 0.45$$

16. (From problem) "Particle's Potential Energy"

$$U(r) = k(x^2 + y^2 + z^2)$$

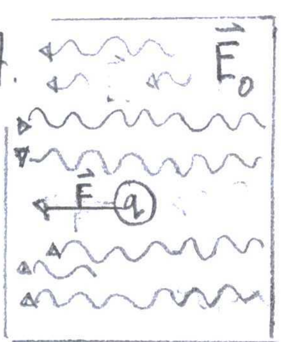
(4.33) "Force on Particle"

$$\vec{F} = -\nabla U$$

$$= -\nabla\left[k(x^2 + y^2 + z^2)\right]$$

$$= -k\left[\left(\frac{d}{dx}x^2\right)\hat{x} + \left(\frac{d}{dy}y^2\right)\hat{y} + \left(\frac{d}{dz}z^2\right)\hat{z}\right]$$

$$= -2k(x\hat{x} + y\hat{y} + z\hat{z})$$

17. 

"charge in uniform electric Field"

(From problem) "Force"

$$\vec{F} = q \cdot \vec{E}_0$$

a) <u>Force is conservative:</u>

Requirements: (i) Only position dependent
(ii) Work is the same for all paths

(4.10) "Work"

$$W(1 \to 2) = \int_1^2 \vec{F} \cdot d\vec{r}$$

$$= \int_1^2 q\vec{E}_0 \, d\vec{r}$$

$$= q\vec{F}_0 \circ \Delta\vec{r}$$

Potential Energy:

(4.13) Potential energy at an arbitrary point

$$U(\vec{r}) = -W(r_0 \to r) \equiv -\int_{r_0}^{r} \vec{F}(r') \circ d\vec{r}'$$

$$= -q\vec{E}_0 \circ \Delta\vec{r}$$

$$= -q\vec{E}_0 \circ (\vec{r} - 0)$$

$$= -q\vec{E}_0 \circ \vec{r}$$

b) Derivative from Potential:

$$F = -\frac{d}{dr} qE_0\vec{r}$$

$$= -\nabla U$$

18. (4.35) "Differential"

$$df = \nabla\vec{f} \circ d\vec{r}$$

a) Perpendicular differential at a point is Normal:

$$y = n \frac{-1}{F'(r)} \circ \Delta r$$

If f is constant at a point, then $df = 0$ because flat surface.

b) <u>df is greatest toward $\vec{r}$:</u>

$$df = \nabla f \circ dr$$

$$= |\nabla f||dr| \cos\theta$$

$$= \underbrace{|\nabla f||dr|}_{\text{max}} \text{ when } \theta^\circ = 0 \text{ toward } r.$$

19 a) (from problem) "Surface"

$$f = \text{const}$$

$$= x^2 + 4y^2$$

The surface is parabolic along the x-dimension and y-dimension, but steeper.

b) <u>Unit normal to surface:</u>

(from problem) "Point"

$$f = 5 \text{ at } (1,1,1)$$

$$\vec{n} = \frac{\langle a,b,c \rangle}{\sqrt{a^2+b^2+c^2}} \quad \text{where} \quad f = ax + by + cz + d = 0$$

$$= \frac{\langle 1,4,0 \rangle}{\sqrt{17}}$$

<u>Maximum rate of change:</u>

$$\frac{\nabla \vec{n}}{|\nabla \vec{n}|} = \frac{\left\langle \frac{\partial n}{\partial x}, \frac{\partial n}{\partial y}, \frac{\partial n}{\partial z} \right\rangle}{\sqrt{\left(\frac{\partial n}{\partial x}\right)^2 + \left(\frac{\partial n}{\partial y}\right)^2 + \left(\frac{\partial n}{\partial z}\right)^2}}$$

$$= \frac{\langle 1,4,0 \rangle}{\sqrt{17}}$$

$$= \vec{n}$$

20. <u>(Back cover) "Curl in Cartesian":</u>

$$\nabla \times A = \hat{x}\left(\frac{\partial}{\partial y} A_z - \frac{\partial}{\partial z} A_y\right)$$
$$+ \hat{y}\left(\frac{\partial}{\partial z} A_x - \frac{\partial}{\partial x} A_z\right)$$
$$+ \hat{z}\left(\frac{\partial}{\partial x} A_y - \frac{\partial}{\partial y} A_x\right)$$

a) (from exercise) "Force"

$$\vec{F} = k\vec{r}$$

$$\nabla \times F = k\left(\hat{x}\left(\frac{\partial}{\partial y} r_z - \frac{\partial}{\partial z} r_y\right) + \hat{y}\left(\frac{\partial}{\partial z} r_x - \frac{\partial}{\partial x} r_z\right) + \hat{z}\left(\frac{\partial}{\partial x} r_y - \frac{\partial}{\partial y} r_x\right)\right.$$

b) (from exercise) "Force"

$$F = (Ax, By^2, Cz^3)$$

$$\nabla \times F = \hat{x}\left(\frac{\partial}{\partial y}(Cz^3) - \frac{\partial}{\partial z}(By^2)\right)$$
$$+ \hat{y}\left(\frac{\partial}{\partial z}(Ax) - \frac{\partial}{\partial x}(Cz^3)\right)$$
$$+ \hat{z}\left(\frac{\partial}{\partial x}(By^2) - \frac{\partial}{\partial y}(Ax)\right)$$
$$= 0$$

c) (from exercise) "Force"

$$F = (Ay^2, Bx, Cz)$$

$$\nabla \times F = \hat{x}\left(\frac{\partial}{\partial y}(Cz) - \frac{\partial}{\partial z}(Bx)\right)$$
$$+ \hat{y}\left(\frac{\partial}{\partial z}(Ay^2) - \frac{\partial}{\partial x}(Cz)\right)$$
$$+ \hat{z}\left(\frac{\partial}{\partial x}(Bx) - \frac{\partial}{\partial y}(Ay^2)\right)$$
$$= (B - 2Ay)\,\hat{z}$$

21. Gravitational Force is Conservative:

(from problem) "Gravitational Force"

$$F = -\frac{GMm}{r^2}\hat{r}$$

(4.54) "Potential Energy"

$$U(x) = -\int_{x_0}^{x} F(x')\,dx$$

$$= -\int_{r_1}^{r_2} \frac{GMm}{r^2}\hat{r}\,dr$$

$$= GMm\left[\frac{1}{r_2} - \frac{1}{r_1}\right]\hat{r}$$

By another route,

$$U_2(x) = -\int_{r_2}^{r_1} \frac{GMm}{r^2}\hat{r}\,dr$$

$$= GMm\left[\frac{1}{r_1} - \frac{1}{r_2}\right]$$

$$U_{TOT} = U_1(x) + U_2(x)$$

$$= 0 \text{, conservative}$$

22. <u>Coulombs Force is Conservative:</u>

(From problem) "Coulomb Force"

$$F = \frac{\gamma}{r^2}\hat{r}$$

(Back cover) "Curl"

$$\nabla \times A = \hat{\rho}\left[\frac{1}{\rho}\frac{\partial}{\partial\phi}A_z - \frac{\partial}{\partial z}A_\phi\right] + \hat{\phi}\left[\frac{\partial}{\partial z}A_\rho - \frac{\partial}{\partial\rho}A_z\right]$$
$$+ \hat{z}\frac{1}{\rho}\left[\frac{\partial}{\partial\rho}(\rho A_\phi) - \frac{\partial}{\partial\phi}A_\rho\right]$$

$$\nabla \times F = \hat{\rho}\left[\frac{1}{\rho}\frac{\partial}{\partial \phi}(0) - \frac{\partial}{\partial z}(0)\right] + \hat{\Phi}\left[\frac{\partial}{\partial z}\left(\frac{\gamma}{r^2}\right) - \frac{\partial}{\partial \rho}(0)\right]$$

$$+ \hat{z}\frac{1}{\rho}\left[\frac{\partial}{\partial \rho}(\rho \cdot 0) - \frac{\partial}{\partial \phi}\left(\frac{\gamma}{r^2}\right)\right]$$

$$= 0$$

23.

a) <u>The Force is conservative</u>

(from problem) $F = k(x, 2y, 3z)$

$$\nabla \times F = \left[\hat{x}\left(\frac{\partial}{\partial y}(x) - \frac{\partial}{\partial z}(2y)\right) + \hat{y}\left(\frac{\partial}{\partial z}(x) - \frac{\partial}{\partial x}(3z)\right) + \hat{z}\left(\frac{\partial}{\partial x}(2y) - \frac{\partial}{\partial y}(x)\right)\right]k$$

$$= 0$$

b) <u>The Force is conservative</u>

(from problem) $F = k(y, x, 0)$

$$\nabla \times F = k\left[\hat{x}\left(\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(x)\right) + \hat{y}\left(\frac{\partial}{\partial z}(y) - \frac{\partial}{\partial x}(0)\right) + \hat{z}\left(\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(y)\right)\right]$$

$= k[0]$

$= 0$

c) The force is not conservative:

(from problem) $F = k(-y, x, 0)$

$$\nabla \times F = k\left[\hat{x}\left(\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(x)\right) + \hat{y}\left(\frac{\partial}{\partial z}(-y) - \frac{\partial}{\partial x}(0)\right) + \hat{z}\left(\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y)\right)\right]$$

$$= k[2\hat{z}]$$

$$= k(0,0,2)$$

Potential Energies:

a) $U = -\int F dV$

$$= -\frac{k}{2}(x^2, 2y^2, 3z^2)$$

$F = -\nabla U$

b) $U = -\int F dV$

$= -k(yx, xy, 0)$

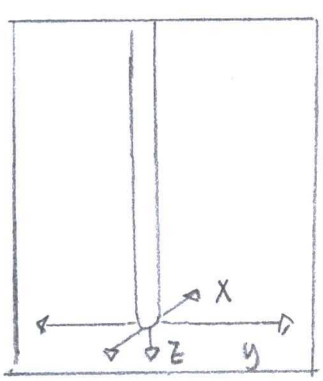
$F = -\nabla U$

c) $U = -\int F dV$

$= -k(-yx, xy, 0)$

$F = -\nabla U$

24.



"infinitely long, uniform rod"

a) Gravitational Force:

(Problem 4.21) "Gravitational Force"

$$F = -\frac{GMm\hat{r}}{r^2}$$

$$= -\frac{GMm\hat{r}}{\rho^2}$$ where ρ = distance from z-axis

$$= -\frac{GM\mu\hat{\rho}}{\rho}$$ μ = mass/length

b) Spherical to rectangular coordinates:

$z = \rho\cos\phi$ at angle $\phi = 0°$ from z-axis.

$$F = -\frac{GM\mu}{z}\hat{z}$$

The Force is conservative:

$$\nabla \times F = -GM\mu\left[\hat{x}\left(\frac{\partial}{\partial y}\left(\frac{1}{z}\right) - \frac{\partial}{\partial z}(0)\right.\right.$$

$$+\hat{y}\left(\frac{\partial}{\partial z}(0) - \frac{\partial}{\partial x}\left(\frac{1}{z}\right)\right)$$

$$\left.+\hat{z}\left(\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial y}(0)\right)\right]$$

$$= 0$$

c) The force is conservative in spherical coord.

$$\nabla \times F = -GM\mu\left[\hat{\rho}\left(\frac{1}{\rho}\frac{\partial}{\partial \phi}\left(\frac{1}{\rho}\right) - \frac{\partial}{\partial z}(0)\right)\right.$$

$$+\hat{\phi}\left(\frac{\partial}{\partial z}(0) - \frac{\partial}{\partial \rho}\left(\frac{1}{z}\right)\right)$$

$$\left.+\hat{z}\left(\frac{1}{\rho}\right)\left[\frac{\partial}{\partial \rho}(\rho \cdot 0) - \frac{\partial}{\partial \phi}(0)\right]\right]$$

$$= 0$$

d) Potential energy:

(14.54) "Potential Energy"

$$U(x) = -\int_{x_0}^{x} F(x')\,dx$$

$$U(x,y,z) = -\int_{x_0}^{x} F(x')\,dx - \int_{y_0}^{y} F(y)\,dy - \int_{z_0}^{z} F(z)\,dz$$

$$= -GMm\left[\int_{x_0}^{x} 0\cdot dx + \int_{y_0}^{y} 0\cdot dy + \int_{0}^{0}\left(\frac{1}{z^2}\right)dz\right]$$

$$= +\frac{GMm}{z}$$

$$= GM\mu$$

25. a) (from problem) "Closed Path"

$$\oint_{\Gamma} F\cdot dr = \int_{A}^{B} F\cdot dr + \int_{B}^{A} F\cdot dr$$

$$= F[B-A] + F[A-B]$$

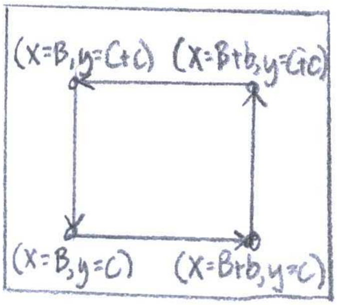
$$= 0$$

b) (from problem) "Stokes Theorem"

$$\oint_{\Gamma} F\cdot dr = \int (\nabla \times F)\cdot \hat{n}\,dA$$

$$= \int (\emptyset) \cdot \hat{n}\, dA$$

$$= \emptyset$$



"Stokes Theorem"

c) (from Problem) "Stokes Theorem"

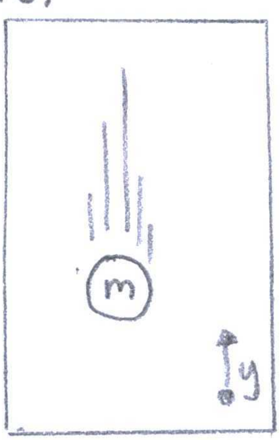
$$\oint_{\Gamma} \vec{F} \cdot dr = \int (\nabla \times \vec{F}) \cdot \hat{n}\, dA$$

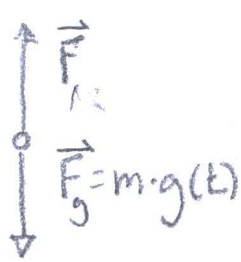
$$= \int_{x=B,\, y=C}^{(x=B+b,\, y=C)} \vec{F} \cdot dr + \int_{x=B+b,\, y=C}^{x=B+b,\, y=C+c} \vec{F} \cdot dr$$

$$+ \int_{x=B+b,\, y=C+c}^{x=B,\, y=C+c} \vec{F} \cdot dr + \int_{x=B,\, y=C+c}^{x=B,\, y=C} \vec{F} \cdot dr$$

$$= 0$$

26. Free-body Diagram





Potential Energy to Force:

(from problem) "Potential"

$U = mg(t) \cdot y$

$\vec{F} = -\nabla U$

$= -\left[\hat{x}\frac{\partial U}{\partial x} + \hat{y}\frac{\partial U}{\partial y} + \hat{z}\frac{\partial U}{\partial z}\right]$

$= -mg(t)$

Energy conserving:

(from problem) "Energy"

$E = \frac{1}{2}mv^2 + U$

$= \frac{1}{2}mv^2 + mg(t) \cdot y$

$\frac{dE}{dt} = m\frac{dg(t)}{dt} \cdot y$

$\neq 0$

27. (4.48) "Time-dependent Potential Energy"

$U(\vec{r},t) = -\int_{r_0}^{r} \vec{F}(\vec{r}',t) \cdot d\vec{r}'$

$$F(r,t) = -\nabla U(r,t)$$

$$= +\nabla \int_{r_0}^{r} \vec{F}(\vec{r},t) \circ d\vec{r}'$$

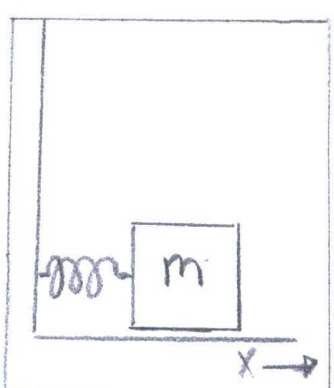
$$= \frac{d}{dr} \int_{r_0}^{r} \vec{F}(\vec{r},t) \circ d\vec{r}'$$

$$= \vec{F}(\vec{r},t)$$

A dilemma was in equations before 4.19. Because time, math from equation 4.13 would have greater complexity; integration variables, formulae, and diagrams.

28. (from problem) "Potential Energy"

$$U = \frac{1}{2} k x^2$$



"mass...on spring"

a) <u>Equation for Conservation of Energy:</u>

(4.20) "Mechanical Energy"

$$E = T + U$$

$$\frac{1}{2}m\dot{x}^2 = E - \frac{1}{2}kx^2$$

$$\dot{x}^2 = \frac{2}{m}\left[E - \frac{1}{2}kx^2\right]$$

$$\dot{x} = \sqrt{\frac{2}{m}\left[E - \frac{1}{2}kx^2\right]}$$

b) (from problem) "Expression"

$$E = \frac{1}{2}kA^2$$

$$\dot{x} = \sqrt{\frac{k}{m}\left[A^2 - x^2\right]}$$

(4.58) "t as a function of x"

$$t = \int_{x_0}^{x} \frac{dx'}{\dot{x}(x')}$$

$$= \int_{x_0}^{x} \frac{dx}{\sqrt{\frac{k}{m}\left[A^2 - x^2\right]}}$$

$$\boxed{\begin{array}{l}\underline{\text{Integral Formula}} \\ \displaystyle\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C\end{array}}$$

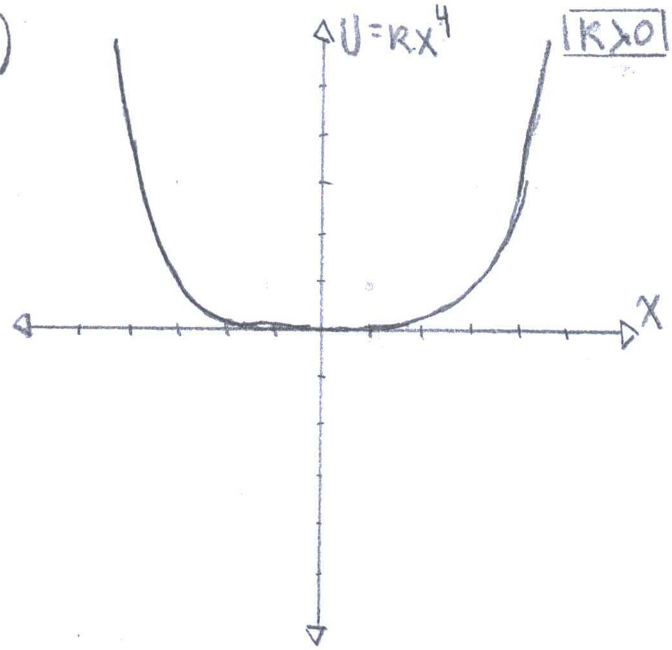
$$= \frac{1}{\omega}\arcsin\left(\frac{x}{A}\right)$$

c) $\underline{x \text{ as a function of } t}$:

$$x = A\sin(\omega t)$$

If $t = 2\pi/\sqrt{m/k}$, then x is simple harmonic motion.

29. a)



At $x=0$, energy is only kinetic energy, but eventually potential by displacement. The potential increases by quartic function. dependent on x-position

b) (4.58) "t as a function of x"

$$t = \int_{x_0}^{x} \frac{dx'}{\dot{x}(x')} = \sqrt{\frac{m}{2}} \int_{x_0}^{x} \frac{dx'}{\sqrt{E - U(x')}}$$

$$= \int_{0}^{x_{max}=A} \frac{dx'}{\sqrt{\frac{2k}{m}\left[A^4 - x^4\right]}}$$

$$= \sqrt{\frac{m}{2k}} \int_{0}^{x_{max}=A} \frac{dx'}{\sqrt{A^4 - x^4}}$$

Period is $\tau = 2\pi\sqrt{m/k}$, so $\tau = 4\pi \cdot t$

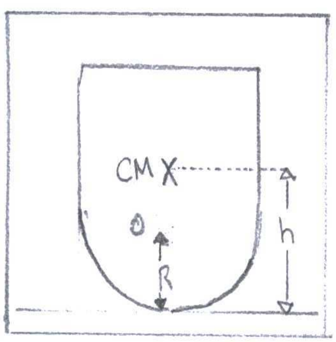
c) $\tau = 2\pi\sqrt{\frac{m}{k}}\int_0^{x_{max}} \frac{dx}{\sqrt{A^4 - x^4}}$

$\propto \frac{1}{A^2}$

d) If $m = k = A = 1$, then $t = \sqrt{\frac{1}{2}}\int_0^1 \frac{dx}{\sqrt{1-x^4}}$

$= 0.93$

30.



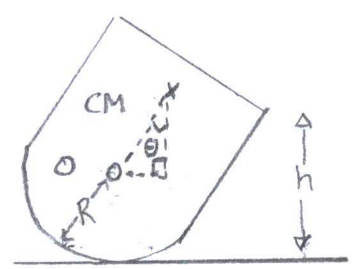
"Child's toy"

a) (pg 115) "Potential Energy"

$U = mgy$

$= mgh$

$= mg(R + (h-R)\cos\theta)$ if



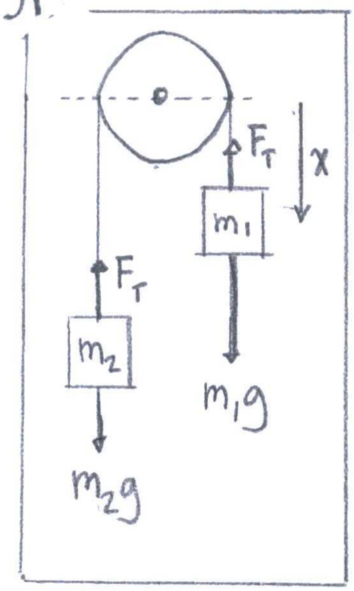
b) Potential has a minimum when $\frac{\partial^2 U}{\partial^2 \theta} > 0$.

Such as $\frac{d^2 U(\theta=0)}{d^2\theta} = +mg(R-h)\cos\theta$,

in a positive range e.g. $R > h$.

But $R > h$ is not reality, so always unstable.

31.



"Atwood machine"

a) (4.64) "Total Mechanical Energy"

$$E = T_1 + U_1 + T_2 + U_2$$

$$= \frac{1}{2}(m_1 + m_2)V^2 + (m_1 + m_2)gh$$

$$= \frac{1}{2}(m_1 + m_2)\dot{x}^2 + (m_1 + m_2)gh$$

b) <u>Equation of Motion:</u>

$$\frac{d}{dx}E = \frac{d}{dx}\left[\frac{1}{2}(m_1 + m_2)\dot{x}^2 + (m_1 + m_2)gh\right]$$

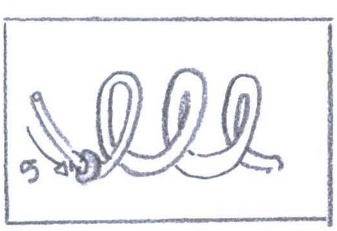
$$= (m_1 + m_2)\ddot{x}$$

$$= M_{TOT} \cdot a$$

$$= F_{TOT}$$

$$= 0$$

32.



"Bead... on a curved rigid wire."

a) <u>Speed:</u>

$$\text{Velocity} = \frac{dx}{dt} = \frac{ds}{dt} = \dot{s}$$

b) <u>Force:</u>

(4.64) "Total Mechanical Energy"

$$E = T + U$$

$$= \frac{1}{2}m\dot{s}^2 + 0$$

$$\frac{dE}{ds} = m\ddot{s}$$

$$= F_{tan}$$

c) Potential derived:

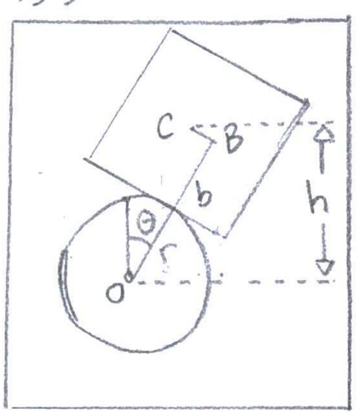
(4.54) "Potential Energy"

$$U(X) = -\int_{X_0}^{X} F_x(x') \cdot dx'$$

$$= -\int_{s_0}^{s} F_{tan} \cdot ds$$

$$F_{tan} = -\frac{dU}{ds}$$

4.33

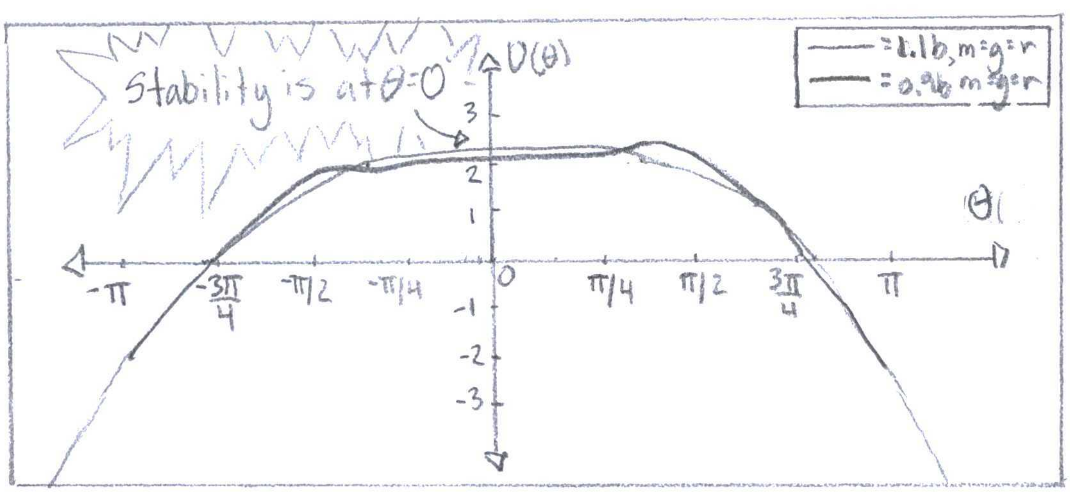


"Cube...placed on cylinder"

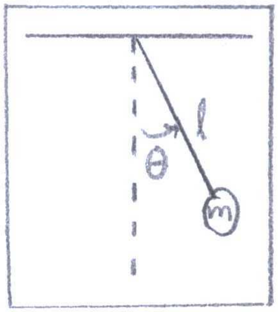
a) (4.59) "Potential Energy"

$$U(\theta) = mgh = mg[\underbrace{(r+b)\cos\theta}_{\text{"Distance from pivot"}} + \underbrace{r\theta\sin\theta}_{\text{"Arc distance rolled"}}]$$

b,c)



4.34.



a) Potential Energy:

$$U(\phi) = U_{final} - U_{initial}$$

$$= mg \cdot l - mgl\cos\phi$$

$$= mgl(1-\cos\phi)$$

Total Energy:

(4.22) "Principle Conservation of Energy"

$$E \equiv T + U$$

$$= \frac{1}{2} I \dot{\phi}^2 + mgl(1-\cos\phi)$$

b) Equation of Motion:

$$\frac{dE}{dt} = \frac{d}{dt}\left(\frac{1}{2} I \dot{\phi}^2 + mgl(1-\cos\phi)\right)$$

$$= I\ddot{\phi} + mgl\sin\phi$$

$$= I\alpha + mgl\sin\phi$$

$$= \Gamma + mgl\sin\phi$$

c) Small angle Approximation: $\phi < 10°$, so $\sin\phi = \phi$

If $\frac{dE}{dt} = 0$, then $-mgl\sin\phi = I\ddot{\phi}$

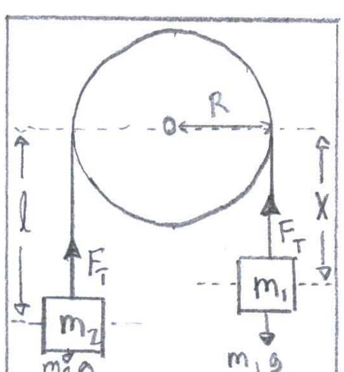
$$-mgl\cdot\phi = ml^2\ddot{\phi}$$

$$\ddot{\phi} + \frac{g}{l}\phi = 0$$

General solution: $\phi(t) = \phi_0 \cos\left(\sqrt{\frac{g}{l}}\,t\right)$

Period (T): $T = 2\pi\sqrt{\frac{g}{l}}$

4.35.



"Atwood machine"

a) Total Energy:

(4.22) "Total mechanical energy"

$$E \equiv T + U \equiv T + U_1(r) + \cdots + U_n(r)$$

$$= T_p + U_p + T_1 + U_1 + T_2 + U_2$$

$$= \frac{1}{2}I\omega^2 + 0 + \frac{1}{2}m_1 v^2 + m_1 g x + \frac{1}{2}m_2 v^2 + m_2 g(l-x$$

$$= \underbrace{\frac{1}{2}\left[\frac{I}{R^2} + m_1 + m_2\right]\dot{x}^2}_{\text{"Kinetic Energy"}} + \underbrace{(m_1 - m_2)gx + m_2 g l}_{\text{"Potential Energy"}}$$

b) Equation of Motion:

$$\dot{E} = \frac{dE}{dX}$$

$$= \frac{d}{dX}\left[\frac{1}{2}\left(\frac{I}{R^2}+m_1+m_2\right)\dot{X}^2 + (m_1-m_2)gX + m_2 g l\right]$$

$$= \left(\frac{I}{R^2}+m_1+m_2\right)\ddot{X} + (m_1-m_2)g$$

$$= 0$$

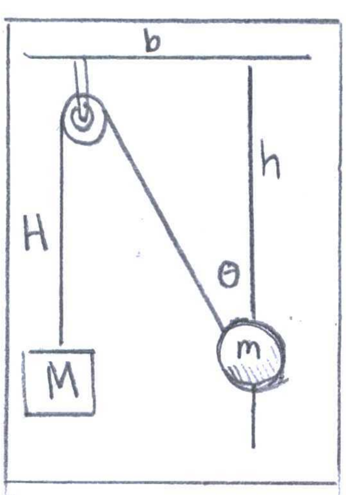
Newton's 2nd Law:

$$F = m \cdot a$$

$$= (m_2-m_1)g$$

$$= \left(\frac{I}{R^2}+m_1+m_2\right)\ddot{X}$$

4.36.



"Metal ball.. on rod"

a) Potential Energy:

(4.59) "Potential Energy"

$$U(\theta) = mgh$$

$$= MgH - mgh$$

$$= MgH - mg\frac{b}{\tan\theta}$$

$$= Mg\left(\frac{l\sin\theta - b}{\sin\theta}\right) - mg\frac{b}{\tan\theta}$$

b) Equilibrium Position:

$$\dot{U}(\theta) = \frac{dU}{d\theta}$$

$$= \frac{d}{d\theta}\left[Mg\left(\frac{l\sin\theta - b}{\sin\theta}\right) - mg\frac{b}{\tan\theta}\right]$$

$$= Mg\left(\frac{b\cos\theta}{\sin^2\theta}\right) - \frac{mgb}{\sin^2\theta}$$

$$= 0$$

$$\theta = \cos^{-1}\left(\frac{m}{M}\right)$$

Why erase??

Identity #1: cot

$$\cot(\theta) = \frac{\cos\theta}{\sin\theta}$$

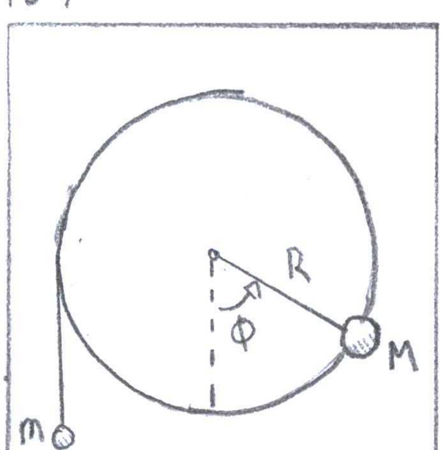
Quotient Rule:

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

The domain requires $\frac{m}{M} < 1$

and a small physical angle by range.

4.37



"mass M.. glued to edge of wheel... and a mass m."

a) Total Potential Energy:

$$U_{TOT} = U_M + U_m$$

$$= (MgR - MgR\cos\phi) - mgR\phi$$

$$= MgR(1-\cos\phi) - mgR\phi$$

b) Positions of Equilibrium:

$$\frac{dU}{d\phi} = \frac{d}{d\phi}\left[MgR(1-\cos\phi) - mgR\phi\right]$$

$$= MgR\sin\phi - mgR$$

$$= 0$$

$$\phi = \sin^{-1}\left(\frac{m}{M}\right)$$

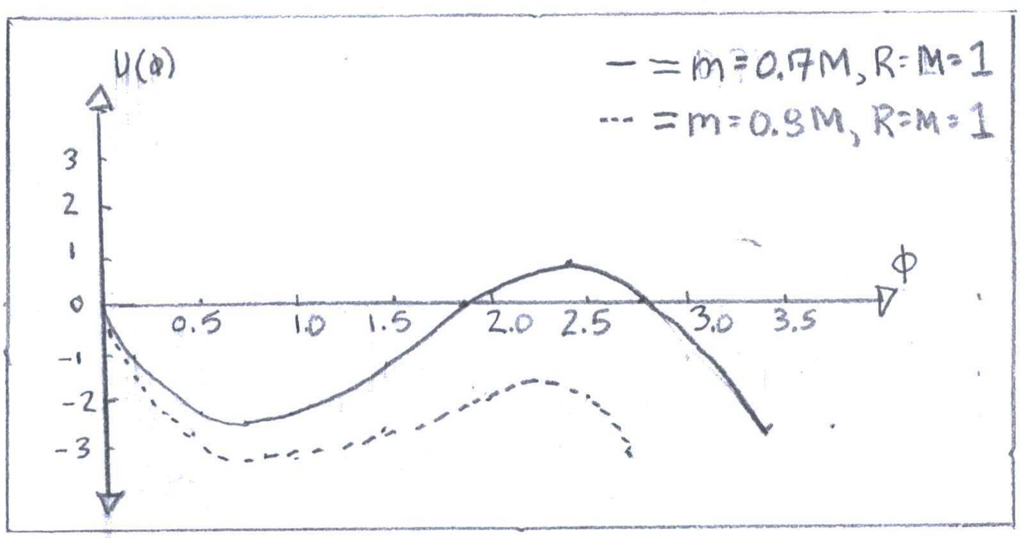
Mass limatations : $0 \leq \frac{m}{M} \leq 1$

Torque Limitations : (pg 90) $T = r \times F$

$$T_M = R \times Mg(1-\cos\phi)$$

$$T_m = R \times Mg\phi$$

The small mass (m) was less than large mass (M) at equilibrium. Additionally the small mass produces more torque.

c) 

d) <u>Critical values:</u>

Reasonably part a, but using Stability $\left(\frac{d^2U}{d\phi^2} > 0\right)$, so $\frac{m}{M} \leq 1$ and $0 < \phi < \frac{\pi}{2}$.

4.38

a) (4.101) "Pendulums Potential Energy"

$$U(\phi) = -m \cdot g \cdot l (1-\cos\phi)$$

Note: Unusual dilemma in problem, so $U(\phi)<0$... now.

$\dot{\phi}$ as a function of ϕ:

(pg 127) "Kinetic Energy"

$$T = \frac{1}{2} m \dot{x}^2$$

$$= \frac{1}{2} m l^2 \dot{\phi}^2$$

(4.22) "Principle Conservation of Energy"

$$E \equiv T + U$$

$$= \frac{1}{2} m l^2 \dot{\phi}^2 - mgl(1-\cos\phi)$$

$$\dot{\phi} = \sqrt{\frac{2(E + mgl(1-\cos\phi))}{ml^2}}$$

Time from $\phi=0$ to $\phi=\max$:

(4.57) "Function of t"

$$t_f - t_i = \int_{x_i}^{x_f} \frac{dx}{\dot{x}}$$

$$t = \int_{\phi_0}^{\phi} \frac{d\phi}{\sqrt{\frac{2(E+mgl(1-\cos\phi))}{ml^2}}}$$

Period (τ):

Trigonometric Identity:

$$\sin^2\left(\frac{\phi}{2}\right) = \frac{1-\cos\phi}{2}$$

$$t = \int_0^{\phi} \frac{d\phi}{\sqrt{\frac{2(E+mgl(1-\cos\phi))}{ml^2}}}$$

$$= \int_0^{\phi} \frac{d\phi}{\sqrt{\frac{2(-mgl(1-\cos\Phi)+mgl(1-\cos\phi))}{ml^2}}}$$

$$= \sqrt{\frac{l}{2g}} \int_0^{\phi} \frac{d\phi}{\sqrt{2(\sin^2(\Phi/2)-\sin^2(\phi/2))}}$$

$$= \frac{\tau_0}{4\pi} \int_0^{\phi} \frac{d\phi}{\sqrt{A^2 - A^2u^2}}$$

$$= \frac{\tau_0}{4\pi} \int_0^{\phi} \frac{d\phi}{A^2\sqrt{1-u^2}}$$

$$\tau = 2t$$

$$= \frac{\tau_0}{2\pi} \int_0^{\phi} \frac{d\phi}{A^2\sqrt{1-u^2}}$$

$$= \frac{\tau_0}{2\pi}\int_0^{\phi} \frac{2\,du}{\cos\left(\frac{\phi}{2}\right)\sqrt{1-u^2}}$$

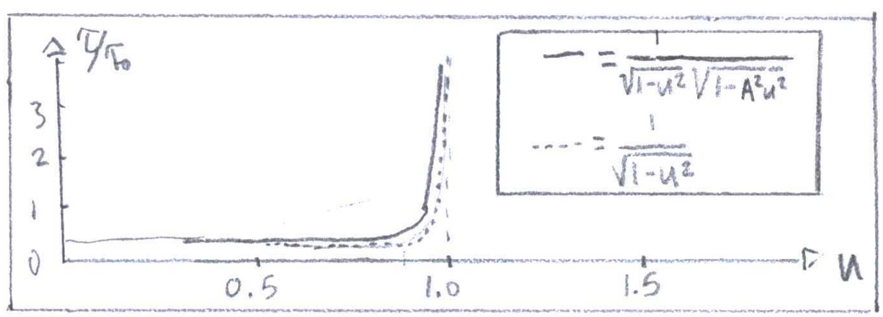
$\boxed{\text{u-sub: } \sin\left(\frac{\phi}{2}\right) = Au}$

$$= \frac{\tau_0}{\pi}\int_0^{\phi} \frac{du}{\sqrt{1-\sin^2\left(\frac{\phi}{2}\right)}\sqrt{1-u^2}}$$

$\boxed{\text{Trigonometric Identity: } 1 = \sin^2\phi + \cos^2\phi}$

$$= \frac{\tau_0}{\pi}\int_0^{1} \frac{du}{\sqrt{1-u^2}\;\sqrt{1-A^2u^2}}$$

b)



4.39 a) Please review Problem 4.38a.

b) The new function is in Problem 4.38b

c) Binomial expansion:

$$\tau = \frac{\tau_0}{\pi}\int_0^{1} \frac{du}{\sqrt{1-u^2}\;\sqrt{1-A^2u^2}}$$

$$\cong \frac{\tau_0}{\pi}\int_0^{1} \left(1+\frac{1}{2}u^2+\cdots\right)\left(1+\frac{1}{2}A^2u^2+\cdots\right)du$$

$$\cong \frac{\tau_0}{\pi}\int_0^1 \left(1+\frac{1}{2}A^2u^2+\frac{1}{2}u^2+\frac{1}{4}A^2u^4\right)du$$

$$\cong \frac{\tau_0}{\pi}\left(\frac{7}{6}+\frac{13}{60}A^2\right)$$

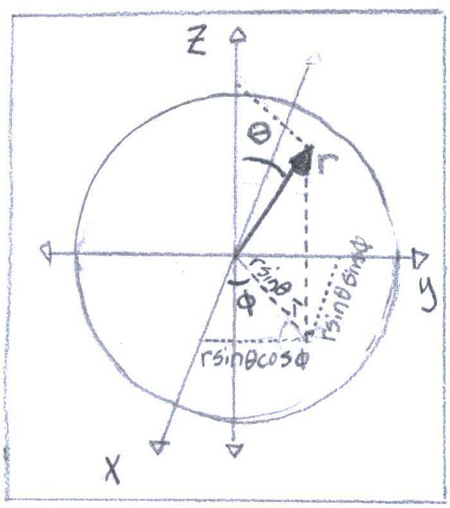
unusual approximation.

$$\cong \frac{\tau_0}{\pi}\left(1+\frac{1}{4}\sin^2(\phi/2)\right)$$

If $\phi = 45°$, then $\tau = \frac{1.036}{\pi}\tau_0$

4.40

a) (4.68) "Spherical Coordinates"



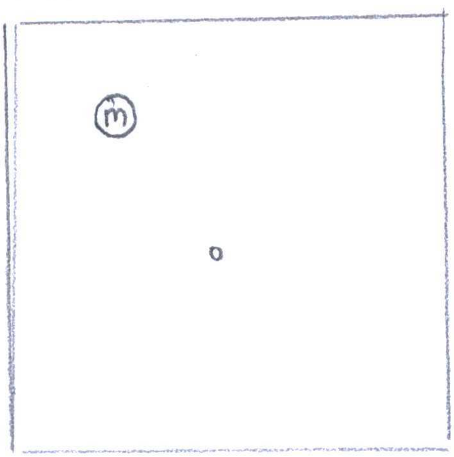
"Cartesian coordinates from spherical"

$$x = r\sin\theta\cos\phi \quad y = r\sin\theta\sin\phi \quad z = r\cos\theta$$

b) $r = \sqrt{x^2+y^2+z^2}$; $\theta = \arctan\left(\frac{\sqrt{x^2+y^2}}{z}\right)$

$$\phi = \arctan\left(\frac{y}{x}\right)$$

4.41



"Mass moves in a circular orbit"

Virial Theorem:

(from problem) "Potential energy"

$$U = k \cdot r^n$$

(4.76) "Spherically symmetric, central forces" - modified

$$-\nabla U = -\hat{r}\frac{\partial U}{\partial r} - \underbrace{\hat{\theta}\frac{1}{r}\frac{\partial U}{\partial \theta}}_{\emptyset} - \underbrace{\hat{\phi}\frac{1}{r\sin\theta}\frac{\partial U}{\partial \phi}}_{\emptyset}$$

If $m_1 \ll m_2$, $\Delta T \cong \frac{(m_2 \cdot V_2^2)}{(m_1 \cdot V_1^2)}$

If $m_1 \gg m_2$ $\Delta T \cong \left(\frac{V_2^2}{V_1^2}\right)$

4.49 (from problem) "Potential energies"

$$U = \gamma \frac{1}{|\vec{r_1} - \vec{r_2}|} \quad \text{where } \gamma = kq_1q_2 \text{ or } -Gm_1m_2$$

Force on particle 1:

$$F_1 = -\nabla_1 U$$

$$= -\left(\frac{\partial}{\partial r_1} + \frac{\partial}{\partial \theta_1} + \frac{\partial}{\partial \phi_1}\right) \gamma \frac{1}{|\vec{r_1} - \vec{r_2}|}$$

$$= \gamma \frac{1}{|\vec{r_1} - \vec{r_2}|^2}$$

Force on Particle 2:

$$F_2 = -\nabla_2 U$$

$$= -\left(\frac{\partial}{\partial r_2} + \frac{\partial}{\partial \phi_2} + \frac{\partial}{\partial \theta_2}\right) \gamma \frac{1}{|r_1 - r_2|}$$

$$= -\gamma \frac{1}{|\vec{r_1} - \vec{r_2}|}$$

(pg 83) "Conservation of Momentum"

$$P = \sum m_\alpha v_\alpha$$

$$P_{in} = P_{fin}$$

$$\sum_{\alpha=1}^{2} m_\alpha v_\alpha = \sum_{\alpha=1}^{2} m_\alpha v_\alpha'$$

$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$

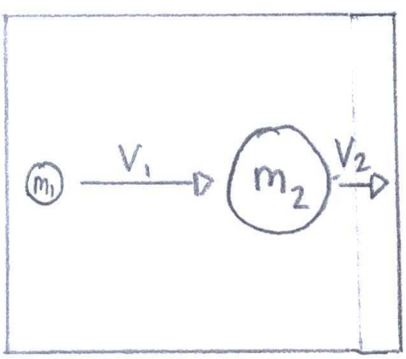
$$m_1(v_1 - v_1') = -m_2(v_2 - v_2')$$

After division (for mass reducing),

$$\frac{m_1(v_1^2 - v_1'^2)}{m_1(v_1 - v_1')} = \frac{-m_2(v_2^2 - v_2'^2)}{-m_2(v_2 - v_2')}$$

$$v_1 - v_2 = -(v_1' - v_2')$$

4.48.



"collision...
inelastic"

Kinetic Energy Changed by fraction:

(4.88) "Conservation of energy"

$$T_{in} = T_{fin}$$

$$\Delta T = T_{fin} / T_i$$

$$= \left(\frac{1}{2}(m_1 + m_2) v_2^2\right) \Big/ \left(\frac{1}{2} m_1 v_1^2\right)$$

$$= V_1'^2 + 2\left(\frac{m_2}{m_1}\right) V_1' \cdot V_2' + \left(\frac{m_2}{m_1}\right)^2 V_2'^2$$

$$= v_1'^2 + 2\left(\frac{m_2}{m_1}\right)|V_1'||V_2'|\cos\theta + \left(\frac{m_2}{m_1}\right)^2 V_2'^2$$

$$= V_1'^2 + \left(\frac{m_2}{m_1}\right) V_2'^2$$

<u>Angle ($\theta < \pi/2$):</u>

$$\theta = \cos^{-1}\left(\frac{\left(1-\frac{m_2}{m_1}\right) V_2'^2}{2|V_1||V_2|}\right) \quad \text{requires} \quad \frac{m_2}{m_1} < 1$$

and $m_2 < m_1$

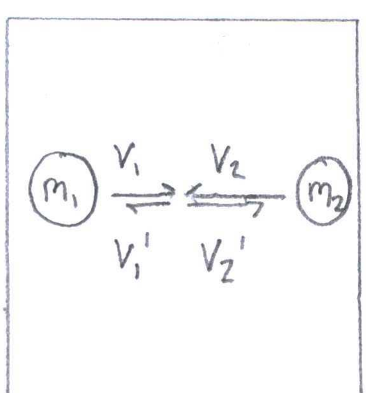
<u>Angle ($\theta > \pi/2$):</u>

$$\theta = \cos^{-1}\left(\frac{\left(1-\frac{m_2}{m_1}\right) V_2'^2}{2|V_1||V_2|}\right) \quad \text{requires} \quad \frac{m_2}{m_1} > 1$$

and $m_2 > m_1$

Facts derived from a plot $\theta = \cos^{-1}((1-x)c)$

4.47. <u>Relative velocity before-and-after:</u>



"head-on collision"

(4.88) "Conservation of Energy"

$$T_{in} = T_{Fi}$$

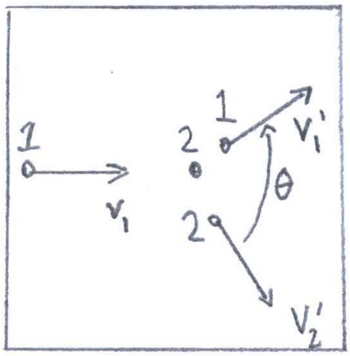
$$\frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 = \frac{1}{2}m_1 v_1'^2 + \frac{1}{2}m_2 v_2'^2$$

$$m_1(v_1^2 - v_1'^2) = -m_2(v_2^2 - v_2'^2)$$

$$= W(A \to B)$$

$$= 0$$

4.46.



"Ellastic Collision"

Angle (θ) derived from unequal masses:

(4.88) Conservation of Energy

$$T_{in} = T_{fin}$$

$$\frac{1}{2} m_1 \vec{v}_1^{\,2} = \frac{1}{2} m_1 \vec{v}_1^{\,\prime 2} + \frac{1}{2} m_2 \vec{v}_2^{\,\prime 2}$$

$$\vec{v}_1^{\,2} = \vec{v}_1^{\,\prime 2} + \left(\frac{m_2}{m_1}\right) \vec{v}_2^{\,\prime 2}$$

(pg 83) Conservation of Momentum

$$P = \sum m_\alpha v_\alpha$$

$$P_{in} = P_{out}$$

$$\sum_{\alpha=1}^{1} m_\alpha \vec{v}_\alpha = \sum_{\alpha=2}^{1} m_\alpha \vec{v}_\alpha^{\,\prime}$$

$$m_1 \vec{v}_1 = m_1 \vec{v}_1^{\,\prime} + m_2 \vec{v}_2^{\,\prime}$$

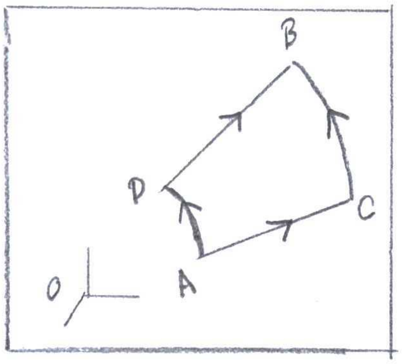
$$\vec{v}_1 = \vec{v}_1^{\,\prime} + \left(\frac{m_2}{m_1}\right) \vec{v}_2^{\,\prime}$$

From energy and momentum,

$$\vec{v}_1^{\,2} = \left(\vec{v}_1^{\,\prime} + \left(\frac{m_2}{m_1}\right) \vec{v}_2^{\,\prime}\right)^2$$

4.44.

(4.7) Work - KE Theorem - modified



"Series of Paths"

$$W(1\to 2) = \int_1^2 F\cdot dr$$

$$W(A\to B) = \int_A^B F\cdot dr$$

$$= \int_A^D F\cdot dr + \int_D^B F\cdot dr$$

$$= \int_A^C F\cdot dr + \int_C^B F\cdot dr$$

If a conservative force, the total work along ADB equals ACB.

4.45

<u>Work by $\vec{F}(r)$ on two paths:</u>

$$W_{ADB} = W_{ACB}$$

$$\int_A^D \vec{F} dr + \int_D^{\vec{B}} \vec{F}\cdot dr = \int_A^C \vec{F}\cdot dr + \int_C^B \vec{F}\cdot dr$$

$$\int_A^D \vec{F}\cdot(r_b - r_a) + \int_D^B \vec{F}(r_b - r_a) = \int_A^C \vec{F}\cdot(r_b - r_a) + \int_C^B \vec{F}(r_b - r_a)$$

$$\underset{=0}{(F(D)-F(A))}\underset{=0}{(r_b-r_a)} + \underset{=0}{(F(B)-F(D))}\underset{=0}{(r_D-r_a)} = \underset{=0}{(F(C)-F(A))}\underset{=0}{(r_b-r_a)} + \underset{=0}{(F(B)-F(C))}\underset{=0}{(r_B-r_a)}$$

b) $\nabla \times F = 0$ (spherical):

(Back Cover) Curl

$$\nabla \times F = \frac{1}{r\sin\theta}\begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ \partial/\partial r & \partial/\partial\theta & \partial/\partial\phi \\ F_r & rF_\theta & r\sin\theta F_\phi \end{vmatrix}$$

$$= \hat{r}\frac{1}{r\sin\theta}\left(\frac{\partial}{\partial\theta}(\sin\theta F_\phi) - \frac{\partial}{\partial\phi}F_\theta\right)$$

$$+ \hat{\theta}\left[\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}F_r - \frac{1}{r}\frac{\partial}{\partial r}(rF_\phi)\right]$$

$$+ \hat{\phi}\frac{1}{r}\left(\frac{\partial}{\partial r}(rF_\theta) - \frac{\partial}{\partial\theta}F_r\right)$$

$$= \hat{r}\frac{1}{r\sin\theta}\left(\frac{\partial}{\partial\theta}(\sin\theta \cdot 0) - \frac{\partial}{\partial\phi}\cdot 0\right)$$

$$+ \hat{\theta}\left(\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}f \cdot r - \frac{1}{r}\frac{\partial}{\partial r}(r \cdot 0)\right)$$

$$+ \hat{\phi}\frac{1}{r}\left(\frac{\partial}{\partial r}(r\cdot 0) - \frac{\partial}{\partial\theta}\cdot f\cdot r\right)$$

$$= 0$$

$$= -k\int_{r_0}^{r} dx + dy + dz$$

$$= -k\left[(r-r_0) + (r-r_0) + (r-r_0)\right]$$

$$= -3k(r-r_0)$$

4.43 a) $\nabla \times F = 0$ (cartesian):

(Back cover) Curl

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_x & F_y & F_z \end{vmatrix}$$

$$= \hat{x}\left(\frac{\partial}{\partial y}F_z - \frac{\partial}{\partial z}F_y\right) + \hat{y}\left(\frac{\partial}{\partial z}F_x - \frac{\partial}{\partial x}F_z\right)$$

$$+ \hat{z}\left(\frac{\partial}{\partial x}F_y - \frac{\partial}{\partial y}F_x\right)$$

$$= \hat{x}\left(\frac{\partial}{\partial y}f\cdot z - \frac{\partial}{\partial z}f\cdot y\right) + \hat{y}\left(\frac{\partial}{\partial z}f\cdot x - \frac{\partial}{\partial x}f\cdot z\right)$$

$$+ \hat{z}\left(\frac{\partial}{\partial x}f\cdot z - \frac{\partial}{\partial y}f\cdot x\right)$$

$$= 0$$

$$= F(r)$$

$$F(r) = -\hat{r}\frac{\partial U}{\partial r}$$

$$= -\hat{r}\frac{\partial}{\partial r}\left[k \cdot r^n\right]$$

$$= -\hat{r} \cdot n \cdot k \cdot r^{n-1}$$

$$= m \cdot a_c$$

$$= -m\frac{v^2}{r}$$

$$= -\frac{2 \cdot \hat{T}}{r}$$

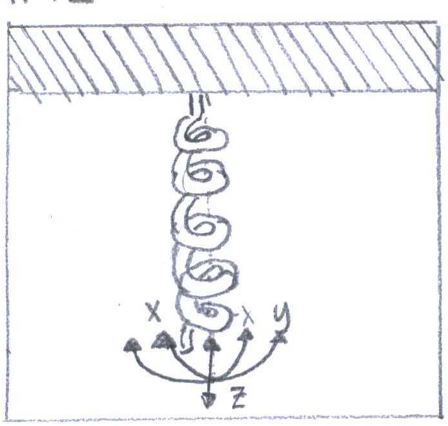
$$\boxed{\underline{\text{Kinetic Energy:}}\quad T = \frac{1}{2}mv^2}$$

$$\hat{T} = \hat{r} \cdot \frac{n}{2} k \cdot r^n$$

$$= \hat{r}\frac{n}{2}U$$

$$T = \frac{n}{2} U$$

4.42



"spring... moves in three-directions"

(From problem) "Hookes Law"

$$F = -kX$$

<u>Force exerted by spring:</u>

$$F = -k\sum_{r=0}^{n} r$$

$$= -k\int_{r}^{r_n} dr$$

4.50. Huzzah! A fact given $\nabla_1 U(r_1 - r_2) = -\nabla_2 U(r_1 - r_2)$.

In one-dimension, $U = f(X_1 - X_2)$

$$-\nabla_1 U' = -f(X_1 - X_2)$$

$$-\nabla_1 U = f(X_1 - X_2)$$

In three-dimensions, the key in Problem 4.49.

4.51 (4.94) "Total Potential Energy"

$$U = U^{int} + U^{ext} = (U_{12} + U_{13} + U_{14} + U_{23} + U_{24} + U_{34})$$
$$+ (U_1^{ext} + U_2^{ext} + U_3^{ext} + U_4^{ext})$$

Potentials:

$$U = U(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_4)$$

$$U^{int} = U^{int}(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_4)$$

$$U^{ext} = U^{ext}(\vec{r}_1, \vec{r}_2, \ldots, \vec{r}_4)$$

$$U_{12} = U_{12}(\vec{r}_1 - \vec{r}_2)$$

$$U_{13} = U_{13}(\vec{r}_1 - \vec{r}_3)$$

$$U_{14} = U_{14}(\vec{r}_1 - \vec{r}_4)$$

$$U_{23} = U_{23}(\vec{r}_2 - \vec{r}_3)$$

$$U_{24} = U_{24}(\vec{r}_2 - \vec{r}_4)$$

$$U_{34} = U_{34}(r_3 - r_4)$$

$$U_1^{ext} = U_1^{ext}(r_1)$$

$$U_2^{ext} = U_2^{ext}(r_2)$$

$$U_3^{ext} = U_3^{ext}(r_3)$$

$$U_4^{ext} = U_4^{ext}(r_4)$$

<u>Net Force on Particle 3:</u>

$$F_3^{Net} = -\nabla_3 U$$

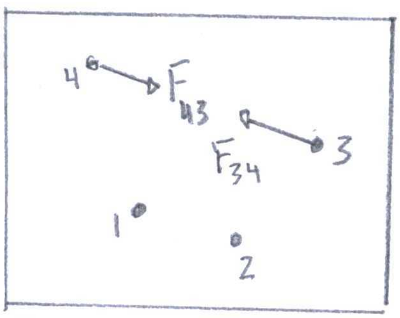
$$= -\nabla_3\left[(U_{13} + U_{23} + U_{34}) + U_3^{ext}\right]$$

4.52.

a) <u>Work-KE theorem:</u>

(4.7) Work-KE theorem

$$\Delta T \equiv T_2 - T_1 = \int_1^2 F \cdot dr \equiv W(1 \to 2)$$



"four-particle system"

$$W_{TOT} = W_{1\to2} + W_{1\to3} + W_{1\to4} + W_{2\to3} + W_{3\to4}$$

$$= \int_1^2 F \cdot dr + \int_1^3 F \cdot dr + \int_1^4 F \cdot dr + \int_2^3 F \cdot dr + \int_3^4 F \cdot dr$$

$$dT = W_{TOT}$$

b) <u>Total Mechanical Energy:</u>

(from problem) $E = T + U$

$$dE = dT + dU$$

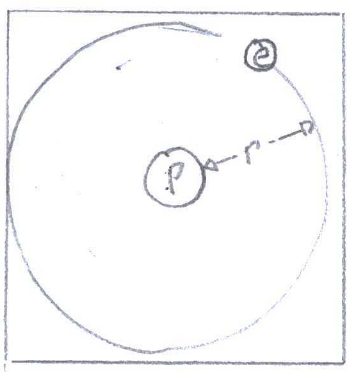
$$= 0 \quad \text{.. when conserved}$$

$$\therefore\ -dU = -dT$$

$$= W_{TOT}$$

$$= -\int_1^2 F\cdot dr - \int_1^3 F\cdot dr - \int_1^4 F\cdot dr - \int_2^3 F\cdot dr - \int_3^4 F\cdot dr$$

4.53



"electron in a circular orbit"

a) Electron Kinetic Energy:

(from Problem 4.41) Potential Energy

$$U = kr^n$$

$$-\frac{\partial U}{\partial r} = -\frac{\partial}{\partial r}(kr^n)$$

$$= -nkr^{n-1}$$

$$= -F$$

$$= -ma_c$$

$$= -m\frac{v^2}{r^2}$$

$$T = \frac{1}{2}mv^2$$

$$= \frac{1}{2}nU$$

b) Total Energy:

$$E = T_e + T_p + V_e + V_p + V_{ep}$$

$$= \frac{1}{2} m_e v_e^2 + \frac{1}{2} m_p v_p^2 - ke^2\left(\frac{1}{r_1} + \frac{1}{r_2} - \frac{1}{r_{12}}\right)$$

c) Values of Total Energy: with another electron

Before: $T_1 = \frac{1}{2}\frac{ke^2}{r_1}$, $T_2 = \frac{1}{2}\frac{ke^2}{r_2}$, $V_1 = \frac{ke^2}{r_1}$

After: $T_1, T_2, V_2 = \frac{ke^2}{r_2}$

KE after ∞-distance is zero.